Book Price Report

Data Scraping, Exploratory Analysis & Predictive Modeling

1. EXECUTIVE SUMMARY

- **Objective**: To architect an end-to-end data pipeline that extracts book catalog data from an online repository, cleans and validates the attributes, performs exploratory data analysis (EDA), and builds a predictive linear regression model for book pricing.

- **Key Findings**: The pipeline successfully collected hundreds of records, handled currency and text transformations cleanly, and established a reliable baseline regression model for evaluating price relationships based on extracted features

Book dataset – 1000

Mean Price = 35.07 pounds

Median Price = 35.98 pounds

Std Deviation = 14.45 pounds

2. Methodology & Architecture

The project follows standard data engineering and machine learning workflows:

1. **Data Extraction**: Automated web crawling via `requests.Session()` and parsing with `BeautifulSoup`. Pagination handling and error checks were embedded to ensure robust data ingestion.
2. **Data Transformation & Hygiene**:
    - Extracted clean numerical floats from raw currency strings (e.g., stripping symbols).
    - Parsed star ratings and stock availability into structured quantitative attributes.
    - Performed automated assertions and checks to guarantee zero null values in critical fields (such as product URLs and titles).
3. **Feature Engineering**:
    - Derived textual features from book metadata, including title length and word counts, to serve as independent variables for modeling.
4. **Predictive Modeling**:
    - Implemented a **Linear Regression** framework utilizing `scikit-learn` to analyze the correlation between structural features and book pricing.

3. DATA COLLECTION & CLEANING METHODOLOGY

Data were collected using Python's requests library and BeautifulSoup. The scraper followed catalogue pagination and extracted book titles, raw prices, textual and numeric ratings, stock status, and relative product URLs.

Price strings were cleaned by removing non-numeric currency and encoding characters and converting the remaining values to numeric prices. Textual ratings were mapped from One–Five to integers 1–5. Stock status was converted into a binary indicator, while title character count and word count were derived as candidate predictors. Original scraped fields were retained for traceability. Validation identified no missing cleaned prices or duplicate product URLs.

EXPLORATORY DATA ANALYSIS & RATING DISTRIBUTION

Descriptive evaluation confirms an overall mean book price of **£35.07** with a standard deviation of **£14.45** and a median price of **£35.98**. Rating groups were relatively balanced, accounting for 17.9% to 22.6% of the catalogue each.

Star Rating Tier	Book Count (n)	Percentage (%)	Mean Price (£)
1 Star	226	22.6%	£34.56
2 Stars	196	19.6%	£34.81
3 Stars	203	20.3%	£34.69
4 Stars	179	17.9%	£36.09
5 Stars	196	19.6%	£35.37
Full Catalog Total	**1,000**	**100.0%**	**£35.07**

EXPLORATORY VISUALIZATIONS

Listed prices ranged from £10.00 to £59.99. The histogram suggests a broadly spread price distribution, while the boxplots show substantial overlap between rating groups.

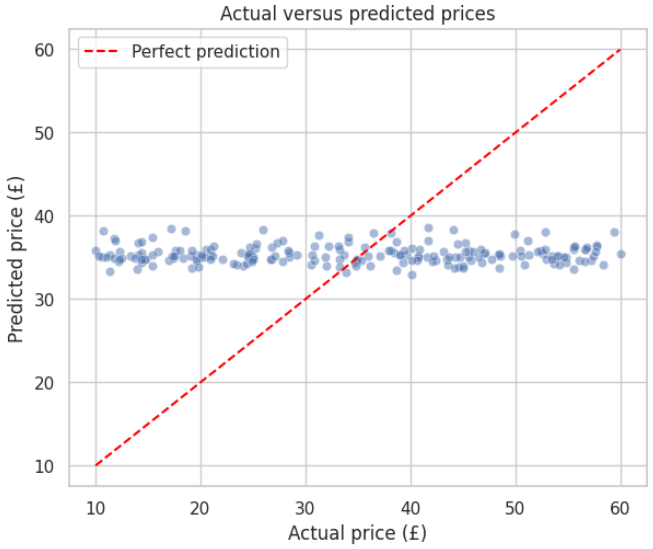


Figure 1: Distribution of listed prices across 1,000 books.



Figure 2: Boxplot showing median and quartile price overlaps across rating tiers.

Mean prices across rating groups ranged from £34.56 to £36.09. The Pearson correlation of approximately +0.0282 indicates a negligible linear association between numeric rating and listed price.

Category and stock-quantity fields were not collected in the current dataset. This limits genre analysis and leaves the brief's stock-quantity requirement incomplete.

4. PREDICTIVE MODELLING & FEATURE JUSTIFICATION

A multiple linear regression model was trained using numeric star rating, title character count, and title word count. Stock availability was excluded because all 1,000 records were marked as in stock. Data were split into 800 training observations and 200 test observations using a fixed random seed of 42. A median-imputation step was included in the modelling pipeline to handle any missing predictor values, with preprocessing fitted only on the training data.

Feature Justification: Rating was included to test whether listed review scores were associated with price. Title length and word count were included as simple text-structure features. These two title measures may capture overlapping information, which limits interpretation of their individual coefficients. Numeric rating encoding assumes equal increments between adjacent star levels.

Model Variant	Test MAE (£)	Test RMSE (£)	Test R^2 Score
Mean-price baseline	£12.66	£14.53	-0.0051
Linear regression	£12.69	£14.58	-0.0113

The baseline predicted the training-set mean price for every test observation. The regression's predictions differed from actual prices by £12.69 on average and performed slightly worse than the mean-price baseline on both MAE and RMSE.

Evaluation Limitations: Evaluation was based on a single train–test split and a limited set of metadata features. The results do not establish statistical independence or rule out other relationships. The demonstration dataset also limits the commercial applicability of the findings.

5. STRATEGIC RECOMMENDATIONS

- **Model is not ready to be deployed:** it did not outperform the mean-price baseline.
- **Complete category and stock-quantity collection:** this addresses missing requirements and enables genre analysis.
- **Evaluate stability using training-set cross-validation:** the current findings are based on one train–test split.
- **Test additional features cautiously:** category and descriptions are candidate predictors, not guaranteed improvements.
- **Use real-market data for commercial decisions:** findings from this demonstration catalogue should not be treated as pricing advice.